Syllabus – ENGR 4010: Senior Capstone Design

Gordon and Jill Bourns College of Engineering Mission Statement and Verse

"Preparing engineering students of excellence and character, with a Christian worldview, who are called to serve, equipped to lead and sent to engage the world with their lives and the appropriate use of technology."

"… For we are God's workmanship, created in Christ Jesus to do good works, which God has prepared in advance for us to do." Ephesians 2:10

COURSE DESCRIPTION

The first of a *two course* senior capstone design sequence. Various design topics may be discussed including equipment design, the design of process systems, and economics. Student teams select a project which may involve company sponsorship, and proceed through the design methodology introduced in earlier design classes, incorporating appropriate engineering standards and multiple realistic constraints. Every project has a 'customer' which requires the generation of a customer spec. During the sequence students provide detailed schedules for building a prototype system or designing a process and present weekly progress reports. They also produce technical specifications, undergo a preliminary design review (PDR), and build a working prototype system if appropriate.

Pre-requisites

ENGR 3020 or EGR 302 – Engineering Design and Documentation (3 units; Spring)

Co-requisites

None

Instructor Contact Information

Dr. Benjamin Sanders, Ed.D., M.S.
bsanders@calbaptist.edu OR 951-552-8251
For emails, put "ENGR 4010" in the subject line
Office: TEGR 335

Office Hours:

TEGR 335, Tuesday: 8:15am – 12:00 noon, 1:15pm – 4:15pm.

Send an email to Dr. Sanders 24 hours prior, if you intend to come to office hours. Otherwise, Dr. Sanders may be working remotely.

Required Text and Resources

Required Text: None

COURSE BIBLIOGRAPHY

- Ian Sommerville, *Software Engineering,* 10th Edition, Pearson Education Limited, 2016.
- Karl E. Wiegers, Software Requirements, 3rd Edition, Pearson Education, 2013.
- Steve McConnell, Code Complete, 2nd Edition, Microsoft Press, 2004.
- Kenneth S. Rubin, Essential Scrum: A Practical Guide to the Most Popular Agile Process, Addison-Wesley Professional, 2012, ISBN-13: 978-0137043293.
- Kent Beck, Extreme Programming Explained: Embrace Change, Addison-Wesley, 2004, ISBN-13: 978-0321278654 (SE)
- Mary Poppendieck, Implementing Lean Software Development: From Concept to Cash, Addison-Wesley Professional, ISBN-13: 978-0321437389.
- C. Dym and P. Little, Engineering Design: A Project-Based Introduction, Third Edition, John Wiley and Sons, 2009.
- G.E. Dieter and L.C. Schmidt, Engineering Design, Fourth Edition, McGraw-Hill, 2009.
- M. Alley, The Craft of Scientific Presentations: Critical Steps to Succeed and Critical Errors to Avoid, Springer-Verlag, New York, 2003.

Purpose of the Course

This team project course includes (typically) 4 students working with the course instructor as 'mentor', a technical advisor, and a client. Development and implementation of a project includes…

- Project Proposal
- Letter of Intent
- Problem Definition
- Client requirements elicitation (interview notes and analysis)
- Background research (literature search, market search, etc. on relevant technology and environment)
- Development methodology
- Risk Assessment
- Prototype Design
- Source Control

Credits and Contact Hours: 3 Credits. 3 contact hours and 6 group hours per week for the duration of the semester.

General Course Objectives

Student Outcomes Covered within EGR 401

- **EGR #7 – Life Long Learning**
  Acquire and apply new knowledge as needed, using appropriate learning strategies.

Course Learning Outcomes

Upon successful completion of this course, students should have learned and understand the concept(s) in each stated outcome.

1. How to define and solve engineering/design problems identified through basic problem statements, client interviews, and background research using proven strategic approaches that lead to high quality solutions. (1)
2. How to apply the general principles of engineering design. (2)
3. How to apply the steps in the general engineering design process and explain both their importance and their inter-relationships. (2)
4. How to apply the appropriate analysis techniques and tools in design. (2)

5. Be able to generate comprehensive design documents as part of a design project. (3, 6)
6. How to apply and implement technical project management methodologies. (5, 9)
7. How to work individually as well as in an engineering team. (5)
8. How to communicate orally and in writing with peers, team members, and supervisors. (6)
9. Be able to articulate how the engineering education at CBU prepares the student to consider the broader impacts of engineering design solutions in a global, economic, environmental, and society context. (6, 8)
10. Be able to independently develop and apply new knowledge to a current software and/or engineering problem. (7)

Course Topics:

- Introduction
- Project management
- Team formation
- General engineering design process topics including customer needs analysis, requirements specification, concept generation and selection, and system design
- Application Development Life Cycle
    - Problem Definition (Software Requirements)
    - Analysis (Software Specification)
    - Design
- Mid-semester presentation and peer assessment
- Final presentation and team peer assessment
- Feedback on deliverables
- Team Meetings and Teamwork
- Written compositions covering ethical issues in engineering technology, design, and lessons learned.

Course Overview:

The primary component of this course will be a senior capstone design project where you will work in a team of students (required 3-5 students per team) to plan, design, prototype, test, and document an engineering product. Other components of the course are the following:

- Deliverables
    - Project Proposal
    - Letter of Intent
    - Background Research
    - Objective Analysis
    - Development Methodology
    - Requirements Specification
    - Risk Assessment
    - Development Approach and Tools
    - Prototyping
    - Project Schedule and Budget
    - Final Design Document
- Project Management (Log Sheet)
- History (Blog)
- Oral presentations and Design Notebook
- Ethics paper

- Teamwork and assessment

Technological Support and Communication

This course will be supported by the Canvas system. The site can be accessed by navigating to https://calbaptist.instructure.com/

- Grades will be posted on the Canvas website.
- Check in ***frequently*** for announcements, assignments, and discussions.
- All homework assignments must be submitted through Canvas (unless otherwise stated). I cannot accept homework that is emailed to me.
- Some course material will be submitted through GitHub. For those assignments a link will need to be submitted on Canvas to support grading.
- Course materials (i.e., Videos (or links), Video quizzes, PowerPoint slides, assignments, etc.) will be posted on Canvas.
- You are responsible for all material covered during lectures **AND** assigned reading **AND** material supplied/assigned via Canvas, not just what appears on the PowerPoint slides.
- ***Note:*** when you communicate with your instructor by email, it is best to use the "Email" link in Canvas. **When using email outside of the Canvas context be sure to put "ENGR 4010" in the subject line!**

This course will use GitHub to submit assignments related to the project itself. Students will be expected to provide their GitHub username and associated email address so that they can be added to the classroom.

ASSIGNMENTS AND GRADING

Grading

Importantly, each student will earn their own individual grade in the context of a group project. Here is how each student will be assessed during Fall 2024.

Attendance and Grading

Importantly, your final course grade will be no higher than your individual Attendance Percentage. Note that only class sessions in which you demonstrate engagement will count towards Attendance Percentage. Engagement entails not disrupting the instructor or other students during class time, arriving on-time as recorded by the instructor, staying focused on class material and any class-related material, and staying in-class for the duration of the class session, including lecturing time and laboratory time. Note that arriving after the start of lecture or leaving early at the end of the class session will likely count as a Tardy, at the discretion of the instructor.

The instructor will keep a record of attendance during the class session, exhibiting the following categories: Engaged, Tardy, Absent, and Excused. Simply showing up on-time is not sufficient to achieve Engaged status, because a student may leave early, disrupt the instructor or other students, or otherwise not be engaged during class time. Any disqualifications from Engaged status for a student who is physically present will likely earn the student a Tardy status. In extreme tardiness, such as showing up within the last 5 minutes of lecture, the student will earn an Absent status despite being physically present. Note that Excused is not a replacement for Engaged, and Excused will not count towards the Attendance Percentage, regardless of the excuse*. The purpose of Excused is to allow a student to make up work with no penalty to the grading of that work. Examples of making up work include taking an exam at a later date or receiving an extension on individual homework assignments. Certain forms of assignments and assessments are not easily made up, despite a valid Excused status. At the instructor's discretion, these may or may not be made up for credit by the student. Examples of such difficult to make-up assignments include presentations in front of other classmates,

assignments that require group work, assignments that require note-taking on a live lecture, and others. Students should not expect to receive opportunities to make up such assignments in general.

*Added 8/19/2025: Excused may count towards the Attendance Percentage if the student demonstrates a valid formal documentation, such as a doctor's note, to the instructor in a timely manner as described in the "Excused" section. Approval of an "Excused" counting towards the Attendance Percentage however is at the instructor's discretion.

Attendance Categories
Here are typical examples of the following categories:

Engaged
A student arrives on-time, listens attentively to the lecture, perhaps takes notes as needed, asks thoughtful, on-topic questions in a respectful way, participates with peers in group activities, demonstrates engagement with the assigned work when prompted by the instructor, focuses on his or her own assignment during laboratory times, and leaves the class once the scheduled class time is over, not before.

Tardy
A student arrives after the instructor has started speaking, utilizes an electronic device for browsing off-topic content, interrupts the instructor or other students, refuses to participate in group activities or is slack in doing so, does not demonstrate engagement with the assigned work when prompted by the instructor, refuses to engage with the assigned work during class time, or leaves the class prior to the end of the scheduled class session.

Absent
The student does not physically show up to the class session, or shows up within the last five minutes of class.

Excused
At least two business days prior to class, the student reaches out to the instructor to request an excused absence for an anticipated event. This may include travel, a scheduled medical appointment, or other events. An unexpected event may also qualify as an excused absence, such as a car crash or other emergency on the same day, prior to the lecture. In either event, the student will likely be expected to supply the instructor with formal documentation of the event within two business days of the lecture. Formal documentation demonstrates the claim of the student and the instructor's ability to contact any formal authority involved. For example, a doctor's note to be used as an excuse needs to express in writing that the doctor excuses the student for a duration of time that covers the lecture time period, and needs to exhibit the legible name, signature, address, and phone number of the doctor. The duration of excused time from such a note will be utilized to determine how much (if any) time will be granted to the student to make up missed assignments that are reasonable to make up with no penalty. Importantly, it is the student's responsibility to reach out to the instructor in a timely way if the student expects to make up assignments or assessments upon receiving an Excused status. For example, if a student contacts the instructor within two business days of the excused absence, this is likely deemed timely. If a student waits until the last week of the semester to reach out to the instructor, the student should likely expect to be denied the opportunity to make up the work.

Item	Weight
Individual Student Project Pitch on First Day on Paper	20%
Signed Letter of Intent	20%
End of Semester Individual Student's Project Component Progress on Canvas	40%
Peer Evaluation on Canvas	20%
Additional ABET Content	To be assessed in Spring semester.
Attendance Percentage	(Cap on Total Grade)
TOTAL	**100%**

Artificial Intelligence Policies

My goal is for you (a human) to learn how to program, and for me to be able to properly assess, pass, and ultimately certify humans who can truly program at a certain level of proficiency. The level of proficiency I look for is intended to distinguish those with a Bachelor's degree in Computer Science from those who have not obtained such a Bachelor's degree. For better or worse, Artificial Intelligence chatbots (AIC) now exist, such as ChatGPT, which can quickly provide realistic, unique solutions in response to assignment prompts. You are an adult human, and by enrolling in classes that I instruct or oversee, you are asking for my certification that you are an adult human programmer. Therefore, as an adult human, I treat you with respect: I respect your freedom to utilize AIC's in your own projects and in your own code as much as you deem meaningful. However, I ask regarding projects that pertain to any classes that you would take under my leadership, that you ensure that you are truly learning despite the level of AIC involvement you decide to use while programming. In other words, if you find that incorporating AIC into your workflow helps you learn how to program more effectively, then I encourage you to utilize AIC. However, if utilizing AIC does not help you learn how to program the specific assignments you are presented with, but in fact hinders you from understanding their true workings to the point where you would not be able to program them when you no longer have access to AIC, please consider minimizing the use of AIC in your workflow.

Participation/Academic Environment

This course meets in class for a consecutive three-hour block of time each week. Typically the first 1 hour of class meeting time may be used for the following.

- Announcements of course policies, events, and assignments.
- Discussion on various issues, including feedback on milestone reports.
- Lectures on topics related to design process, project scheduling, presentation, report and miscellaneous requirements, etc.

The rest of the weekly class meeting time is for the instructor to provide feedback on deliverables and for the teams to meet. It should be noted that teams are expected to work on the project for at least six hours per week in addition to the three hours of class time (**a total of nine hours per week**).

ABET - Design Document(s) – For each of the required deliverables (as indicated in the Course Calendar in this syllabus), each team must submit a written report containing all the relevant information for the step in an organized manner. **In addition to being submitted on Blackboard, each design file must also be sent to the advisor and the client.** Evidence that acknowledges the receipt (and suggestions for improvement if any) by the advisor and the client must also be obtained and submitted on Blackboard WITHIN ONE WEEK OF THE DUE DATE OF THE DELIVERABLE. Failure to do so may result in a 50% deduction of the deliverable grade for each missing acknowledgement. (Note: acknowledgment only acknowledges the receipt of the report but does not necessarily imply that the quality of the report meets expectations). Based on the deliverable, the instructor will (as well as the advisor and the client may) offer oral or written feedbacks and suggestions for improvements to the team. The team can then improve the design file deliverable prior to either the midterm and/or

final presentations. The team may also choose to include additional information they deem relevant. It is the team's responsibility to organize the design document(s) in the best, professional manner in a design notebook. **It is important to have version information on the deliverables and keep all major versions of the deliverables (including the acknowledgement and suggestions from the instructor/client/advisor) in the notebook.** The team should also keep an electronic copy of the notebook. The notebook should become a part of your senior portfolio.

Time Logs: EACH STUDENT MUST LOG THEIR INDIVIDUAL TIME SPENT ON THE PROJECT TO THE NEAREST 0.25 HOURS, including a short (one line) description of the task performed. This will be used for (1) understanding engineering economics and the cost of engineer's time, (2) a metric for evaluating team effort, and (3) closing the loop with respect to the project management (i.e. Gantt chart.). **Note that you are required to include each team member's time logs in the design document and you may be required to present and/or submit your time log at any time throughout the semester. It is highly recommended that each student keep a log book throughout the design process. These are to be updated weekly in a Google Sheet.**

Blogs: Teams will be expected to document progress using an online blog. The blog should contain an ongoing summary of work done by the team. This would include summaries of meetings, pictures of white boards from brainstorming sessions, scanned images of hand written notes or designs, etc. The blog should contain information that shows what the team accomplished each week.

ABET - Team Assessments: Team peer assessments will also be conducted in the middle and at the end of the semester, which provide the team members an opportunity to rate each other (see the note in "Grading Policy" for more information).

Waterfall Expectations: The following are some expectations of the Capstone Design projects, if teams are following Waterfall, for this semester.

- By the end of October, teams are expected to have completed a preliminary design and defined a development strategy. At that time all major interfaces should be prototyped to confirm proof of concept on the overall design.
- The team must consider the relevant engineering standards and time, resources, and other realistic constraints in their design.
- Students will be given additional instructions on the requirements for presentations and expected format and organization of the reports.
- At the completion of the Capstone Design sequence, teams are required to prepare a working schedule (including a "burn down" chart) that will be used to manage development within EGR 402.
- ABET - In addition to the capstone project, each student is to write an Ethics Paper (the requirements will be provided by the course instructor). This assignment must be completed to pass the course.
- ***EGR 401 and EGR 402 should be completed in the same academic year. IF students complete EGR 401 but are unable to complete EGR 402 within that same academic year then they will be highly encouraged to take EGR 401 again.***

While we expect and encourage students to consult regularly with their faculty advisors, other faculty, or experts in industry, we feel that for the course to qualify as a "major design experience", the primary responsibility for project progress, scheduling, management, and reporting must lie with the students**. This demands substantial initiative and motivation on the students' part.** Students should not expect the instructor to provide all guidance.

All the presentations and reports, as with all class work, are to be accomplished in a professional and ethical manner. Integrity in all work is expected. Although a team's project results may not always be all that was hoped for, anything

other than straightforward and honest reporting will not be tolerated. This course has been set up to emulate a typical engineering project. As such, it is anticipated that a meaningful learning experience will result.

Final Thoughts: Including composition skills, oral presentation capabilities, project management techniques, and self-determination of development details for a design course is part of the effort to provide CBU Engineering students access to many aspects of what they will encounter in their professional career. This effort is being implemented to better prepare the engineering student to enter the engineering workforce well versed in the CBU School of Engineering Program Objectives and as the ECE/SE Program Outcomes as presented in this syllabus and in support of ABET's (Accreditation Board of Engineering Technology) Engineering Criteria 2000. The Criteria is intended "to provide a foundation of knowledge and skills to students in the non-technical side of engineering by focusing on concepts such as communication, ethics, problem-solving, teambuilding, and other everyday issues outside of the technical aspects of the profession." Therefore, EGR 401 is a component in the CBU Engineering students' graduation path that is intended to help prepare for a smooth entry into the work areas that will be encountered in the engineering field. In other words, EGR 402 is structured to be a "real-world experience" with a Christian foundation.

You are encouraged to commit yourself to a standard of excellence in this course that perhaps you have not yet done in your academic career. This course offers you a wealth of valuable experience. Your attitude will determine the depth to which you will be able to mine it (see Proverbs 13:4 and Proverbs 22:29). Your rewards will be great. May the Lord bless the works of your hands (Psalm 90:17). Let the adventure begin!

ABET - Ethics Paper

Students will write an essay that will identify, provide background for, analyze, and draw conclusions on a computer science or software engineering project failure. Your essay will be graded on both the content and the composition. Each student must individually complete this assignment to pass the course.

ABET - Self-Reflection Paper

At the end of the course students will reflect on things learned, things accomplished, and what could have been done differently to improve the overall process or result. Each student must individually complete this assignment to pass the course.

Program Assessment Graduation Requirement

Students needing to complete the ETS - Major Field Test in Computer Science graduation requirement will do so within EGR401/402 (this will also cover EGR403 for those on older catalogs). The requirement will be clearly defined in a separate document, in Spring semester.

Optional Project Management (weekly Log Sheets)

Each week's schedule and priorities for all team members will be presented to the instructor during the team meeting in class for approval at the beginning of each week. Once approved all tasks should be entered into the team log sheet. Completion status and hours should be updated throughout the week. At the end of each week the team will be graded according to the following formula:

Hours Log/Plan * Progress

Hours Log/Plan

- 100% if posted by 8 AM on the day class each week
- 80% if posted before the end of class
- 70% if posted 24 hours late
- -10% for each additional 24 hours after that

Progress:

- 100% if weekly goals were accomplished by 8 AM on the day of class each week
- 80% if weekly goals were accomplished before the end of class
- 70% if accomplished 24 hours late
- -10% for each additional 24 hours after that

Optional Weekly Blog Posts

A history of all team activity and accomplishments will be published via a weekly blog. At the end of the week the team blog will be graded according to the following formula:

On Time Factor * Quality Factor

On Time Factor

- 100% if posted by 8 AM on the day of class each week
- 80% if posted after class starts but before the end of class
- 70% if posted 24 hours late
- -10% for each additional 24 hours after that

Quality Factor:

- 100% if all accomplished goals are clearly demonstrated
- 80% if most accomplished goals are demonstrated
- 60% if some accomplished goals are demonstrated

Optional Meeting Minutes

The final design document should have an appendix with minutes from ALL team meetings. This applies to all meetings whether all members are present or not. This also applies to all meetings with clients and/or technical advisors. Minutes should reflect the date/time of the meeting, location, who was present, and what was discussed. Action items should also be specified. The first action in any team meeting would be the review and approval of the minutes from the previous team meeting.

ABET - Peer Assessments

Team members will provide an assessment of all team members (including themselves) twice during the semester: at the midterm point and after the final presentation. It is HIGHLY recommended that each student thoughtfully review themselves and their peers. There are situations where team members should earn a perfect score, but those types of team members are not common and if you choose to give someone a perfect score then you should be willing to comment on their outstanding performance. Comments that summarize team dynamics and other factors that may have contributed to why a team member was scored a certain way are highly encouraged (and might ultimately be required to get full points for this assignment). Peer assessments must be submitted to pass the course.

Late Policy

Any assignments which are submitted late (after the due date on Canvas) without prior approval from the instructor will be penalized 100% (i.e., the grade will be a zero). You must still submit all assignments to pass the course.

Make-Up Policy

Makeups for missed meetings or presentations will be allowed in the event of an illness or emergency, but the instructor must be notified and approve it before the meeting or presentation (in the case of an emergency, notify the instructor within 24 hours after the meeting or presentation takes place). In all cases, the student must show evidence from a doctor, police, or other relevant agencies as proof of illness or emergency. **There will be no makeups for unexcused absences or absences notified after 24 hours of the meeting or presentation.**

If a student needs to miss a meeting or presentation for an official, unavoidable university event (e.g. athletic competition or choir performance), the instructor needs to be notified by a university official prior to the event. The student is responsible for contacting the instructor and making arrangements for a make-up at least one week prior to the meeting or presentation. There will be no makeups if the student does not communicate with the instructor and make arrangements at least one week prior to the meeting or presentation.

Final Grades

The following scale will be used to calculate final grades:

A	93%-100%	A-	90%-92.9%	B+	87%-89.9%
B	83%-86.9%	B-	80%-82.9%	C+	77%-79.9%
C	73%-76.9%	C-	70%-72.9%	D+	67%-69.9%
D	63%-66.9%	D-	60%-62.9%	F	< 59.95%

Checking Progress and Grades

Course resources and grades will be posted on Blackboard. Please check it on a regular basis. In addition, you are expected to check your CBU e-mail at least once every 24 hours during the work week (Mon-Fri).

EXPECTATIONS

Academic Integrity

Discussion of assignments with other students within your own team is generally encouraged, unless specifically prohibited by the instructor. **All work submitted for grading, however, must be your own.** The exams and progress reports are to be individual effort only. There is to be no communication whatsoever about course or exam material during the exam or with individual students who may be taking an exam early or late. No cheating of any kind is permitted or tolerated.

Any evidence of plagiarism, academic dishonesty, or other violations of the CBU Honor Code may constitute grounds for a failing grade in the course and will, as a minimum, lead to no credit for the exam or assignment in question.

The course penalty for violating the Honor Code will be determined by the instructor. Any incident of academic dishonesty (cheating, plagiarism, copying, and other forms) must be reported to the Dean of Students. A first incident of cheating may be handled at the discretion of the instructor in consultation with the Dean. Judicial sanctions may include, but are not limited to, loss of a letter grade or failure in the course in which the offense occurred, suspension, and/or dismissal from the University. A detailed discussion of academic dishonesty appears in the CBU *Student Handbook.* For further guidance please refer to the CBU Student Handbook for the CBU Honor Code and to the College of Engineering Policy on Academic Integrity. **Plagiarism** refers to representing work as your own without giving credit to the original author. Paraphrasing another person's work without citing the author is also plagiarism. For this course turning in work that you completed for another course is unacceptable. Also, turning in work that has been copied from a solution key or another student's assignment is unacceptable. All university policies pertaining to plagiarism will be enforced in this course. As a point of information and clarification, you may wish to visit the following website: http://www.plagiarism.org.

Academic Environment

Students are expected to encourage and maintain a positive learning environment that is reflective of a "Christian community of learners." Actions and behavior that distracts fellow students or is disruptive in the classroom will not be tolerated. To help maintain a positive learning environment, the following policies will be enforced:

- ❑ Cell phone use is prohibited in the classroom, unless specific permission is granted on a case by case basis. Cell phones should be silenced and not in view.
- ❑ Laptops and tablets should be closed and put away unless the instructor authorizes their use for class purposes (no email, Facebook, etc. during class).
- ❑ Earphone use is prohibited in the classroom.
- ❑ Lectures may NOT be recorded without permission.
- ❑ Class members and the course instructor are to be treated with proper respect at all times.
- ❑ At the end of class, all students are expected to clean up after themselves. If desks (or other classroom furniture) was moved during class then they should be returned to the configuration found at the beginning of class.

Inability to follow these rules or other actions that distract students and disrupt the learning environment will lead to a reduction in the "Academic Environment" portion of the grade and could lead to removal from the classroom for that day (and this would count as an unexcused absence).

CSDS Rules of conduct:

As college students and soon-to-be professionals, your conduct around your peers and your professor are expected to be polite, respectful and non-distracting. In particular, this means the following:

- No use of cell phones in class (for calls, texting, browsing, gaming, playing music or any other function) unless required for a classroom activity. Check with your instructor if you have any questions.
- No use of any computer for a non-class related purpose (taking notes is OK, working on homework, gaming or browsing is not)
- Earphones are not allowed during any class sessions.
- Do not record the lecture without permission.
- No talking or other behavior that might distract another student
- No treating another class member or the instructor with disrespect
- No submitting late work via email (or any other method) pleading for consideration

For each violation of any of the above guidelines, five points will be deducted from your overall grade for the course

Late assignments

- Late assignments **WILL NOT BE ACCEPTED**.
- Assignments are submitted via Blackboard through any terminal that is connected to the Internet.
- Students are encouraged to submit assignments early; do not wait until the last possible moment to submit assignments.
- There are no acceptable excuses for the assignments being late.

Drop Form

- If a student decides to drop the course, it is their responsibility to do so by submitting a Drop Form to the registrar's office.
    - The instructor cannot drop a student from a class.
    - If the course is not properly dropped, the student will receive an "F" as the final grade.

Class Preparation

- Students are expected to prepare for each assignment prior to the class which presents the topic.

- There will be in class discussions and demonstrations.
    - Demonstrations are not the sole responsibility of the instructor, but from time to time individuals and teams will be required to demonstrate a technique.
- Plan to spend approximately six hours each week working on reading, reviewing, homework, and laboratory assignments to obtain a grade of "C" in this course.

Academic Dishonesty

- Academic dishonesty, as explained in the student handbook HONOR CODE policy, ***WILL NOT BE TOLERATED***.
- Each student should become familiar with those offenses identified in the student handbook.
- A failing grade in the course will result from offenses identified as "cheating," especially the misrepresentation of assignments.
- **ALL COURSE WORK IS THE SOLE RESPONSIBILITY OF THE STUDENT**. Work performed by a student other than the name appearing on the assignment turned in, will be considered misrepresentation for **both** students for the assignment. At a minimum, a failing grade for the assignment and potentially the course will result from any incident of academic dishonesty.
- Students are expected to uphold the school's standard of conduct relating to academic honesty.
- Students assume full responsibility for the content and integrity of the academic work they submit. The guiding principle of academic integrity shall be that a student's submitted work, examinations, reports, and projects must be that of the student's own work.
- Students shall be guilty of violating the honor code if they:
    1. Represent the work of others as their own.
    2. Use or obtain unauthorized assistance in any academic work.
    3. Give unauthorized assistance to other students.
    4. Modify, without instructor approval, an examination, paper, record, or report for the purpose of obtaining additional credit.
    5. Misrepresent the content of submitted work.
- The penalty for violating the honor code is severe. Any student violating the honor code is subject to receive a failing grade for the course and will be reported to the Office of Student Affairs and Provost. If a student is unclear about whether a particular situation may constitute an honor code violation, the student should meet with the instructor to discuss the situation.

It is permissible to assist classmates in general discussions of computing techniques. General advice and interaction is encouraged. Each person, however, must develop his or her own solutions to the assigned projects, assignments, and tasks. In other words, students may not "share solutions" on graded assignments. Such collaboration constitutes cheating. A student may not use or copy (by any means) another's work (or portions of it) and represent it as his / her own. If help on an assignment is needed, contact the instructor; do not seek solutions from other classmates.

Classroom Behavior

Any acts of classroom disruption that go beyond the normal rights of students to:

- question and discuss the instructor's educational process or outside relative subject content,
- ingress and egress the classroom on time, or
- conduct normal communication

will not be tolerated (see Student Life Policy described in the Student Handbook).

Children in class

The University does not provide childcare for the dependents of students, faculty, or staff. Students are not permitted to be accompanied by children / dependents during class sessions. (NOTE THE CHILDCARE (BABY-SITTING) POLICY IN YOUR STUDENT HANDBOOK.)

APPEALS POLICY

To appeal a grade on an individual assignment, send e-mail to your instructor's e-mail address within two weeks of the grade having been received. Overdue appeals will not be considered. To appeal a final course grade, use the normal CBU appeals process.

INCOMPLETE POLICY

Students will not be given an incomplete grade in the course without sound reason and documented evidence as described in the Student Handbook. In any case, for a student to receive an incomplete, he or she must be passing and must have completed a significant portion of the course.

DISABILITIES POLICY

In compliance with the Americans with Disabilities Act (ADA), all qualified students enrolled in this course are entitled to "reasonable accommodations." Please notify the instructor during the first week of class of any accommodations needed for the course.

TUTOR ASSISTANCE

Tutors are available in the Office of Student Success in Lancer Arms #54.

SEXUAL HARASSMENT & TITLE IX POLICY

Review the Sex Discrimination, Sexual Violence & Sexual Harassment section of the Student Handbook. All offences will be reported.

Useful Links

Student Handbook:
http://www.calbaptist.edu/explore-cbu/offices/office-registrar/academic-catalogs/undergraduate/

Academic Catalog:
http://www.calbaptist.edu/explore-cbu/offices/office-registrar/academic-catalogs/undergraduate/

Calendar:
https://insidecbu.calbaptist.edu/ICS/Academics/Academic_Calendars.jnz

ACADEMIC POLICIES AND PROCEDURES

The academic policies and procedures of California Baptist University are contained in the Academic Catalogs, available at http://catalog.calbaptist.edu/. Academic policies such as independent and directed study courses, academic probation and waivers, grade change, incomplete grades, student account policies and fees are documented in the current academic year catalog. Specific degree requirements depend on your "catalog year"; check with your advisor for more information. Be sure to refer to the "undergraduate" or "graduate" catalog as appropriate for you.

Recording Class Sessions

Recording of class sessions without the prior express written permission of the instructor is prohibited. Any permission granted shall include the requirements that a recording may only be

used for content study purposes only and sharing a recording with anyone outside of the course and/or posting on social media are strictly prohibited. This includes any recordings that the instructor shares with students. This course policy is in alignment with Student Handbook and the Standard of Student Conduct. Refer to Student Handbook policies 15.6, 15.7, and 15.8 for more information.

Calendar

Week #	Section F	Section H	Topic	Due
1	9/9	9/14	Teams Establishment, Project Establishment	Individual Student Project Pitch on Paper
2	9/16	9/21	Work in Teams	
3	9/23	9/28	Work in Teams	
4	9/30	10/5	Work in Teams	
5	10/7	10/12	Work in Teams	
6	10/14	10/19	Work in Teams	
7	10/21	10/26	Work in Teams	
8	10/28	11/2	Work in Teams	
9	11/4	11/9	Work in Teams	
10	11/11	11/16	Work in Teams	
11	11/18	11/30	Work in Teams	
12	12/2	12/7	Work in Teams	Peer Eval on Blackboard
13	12/9		Work in Teams	Individual Project Submission on Blackboard

12 Attendance Percentage Points Possible for Section H.

13 Attendance Percentage Points Possible for Section F.

Student Outcomes (for the College of Engineering)

Student Outcomes (SO's) describe what students are expected to know and be able to do by the time of graduation. These relate to the knowledge, skills, and behaviors that students acquire as they progress through the program. The SO's that are addressed by the Course Outcomes are in parenthesis following the Course Outcomes list. The detail information on the various programs in the CBU Gordon and Jill Bourns College of Engineering can be found on the programs' websites.

Engineering Student Outcomes:

EGR #1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics (ABET)

EGR #2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors (ABET)

EGR #3. an ability to communicate effectively with a range of audiences (ABET)

EGR #4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts (ABET)

EGR #5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives (ABET)

EGR #6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions (ABET)

EGR #7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies. (ABET)

EGR #8: an ability to articulate a Christian worldview on personal, professional, technical, and societal issues

EGR #9: An ability to develop attributes of a leader in an innovation-driven environment